A Machine-Checked Core for Set Theoretic Estimation

Project `thejeb`

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Abstract

We report a first, machine-checked formalization in Lean 4 / Mathlib of the core of set theoretic estimation (STE) as founded by Combettes [3] and recast over topological spaces for decryption by Carlson [2]. The formalization covers the feasibility set and its lattice-theoretic properties, the consistent / fair / ideal taxonomy, information monotonicity, an inconsistency-based invalid-information detector, closedness and compactness-based existence of estimates on topological spaces, convexity of the feasibility set, and a skeleton of Carlson's STE-decryption reduction. Every theorem stated here is verified by continuous integration on each push; nothing is asserted proven otherwise.

1 Introduction

Set theoretic estimation abandons the search for an optimum in favor of *feasibility*: an estimate is acceptable exactly when it is consistent with every available piece of information [3]. This shifts the mathematical content from analysis of an objective function to the set algebra and topology of a family of constraint sets and their intersection. That content is a natural target for a proof assistant. We mechanize it in Lean 4 on Mathlib (pinned `v4.32.0`), adopting a strict discipline: a statement counts as established only when it type-checks in our library under the project's CI. This paper is the human-readable companion to that library; every displayed result names its Lean identifier.

2 The formal model

Let Ξ be the solution space and I an index set of information pieces. A *property set family* is $S : I \to \mathcal{P}(\Xi)$; S_i holds the estimates consistent with information i. We mechanize the crisp case (Combettes's membership function Ψ_i being $\{0,1\}$-valued, so S_i is a subset).

Definition 1 (feasibility set; `STE.feasibilitySet`). *$S = \bigcap_{i \in I} S_i$ (Combettes Eq. 4). A point of S is a* set theoretic estimate*.*

```
def feasibilitySet (S : I -> Set Xi) : Set Xi := iInter i, S i
theorem mem_feasibilitySet : a in feasibilitySet S <-> forall i, a in S i
```

3 Results (all machine-checked)

Refinement and monotonicity. The feasibility set refines each constraint (`feasibilitySet_subset`: $S \subseteq S_j$). Writing $S_J = \bigcap_{i \in J} S_i$ for the *partial* feasibility set, enforcing more information shrinks it:

Proposition 1 (`partialFeasibilitySet_antitone`). *$J \subseteq K \implies S_K \subseteq S_J$.*

This is the STE reading of "more information never enlarges the acceptable set," and it descends to Carlson's key search as `feasibleKeys_antitone`.

Consistency, fairness, invalid information. Call the family *fair* for h when $h \in S_i$ for all i (`STE.Fair`); this is exactly $h \in S$ (`fair_iff_mem_feasibilitySet`).

Theorem 1 (`feasibilitySet_nonempty_of_fair`). *Fair information is consistent: if the truth satisfies every piece of information then $S \neq \emptyset$.*

Contrapositively, inconsistency is a certificate that some information is wrong:

Theorem 2 (`exists_unfair_of_feasibilitySet_eq_empty`). *If $S = \emptyset$ then for every candidate h there is an i with $h \notin S_i$.*

Ideal information. The family is *ideal* for h when $S = \{h\}$ (`STE.Ideal`).

Theorem 3 (`Ideal.eq_of_mem`, `ideal_iff_fair_and_subsingleton`). *Under ideal information every estimate equals the truth; and idealness is equivalent to fairness together with feasibility being a subsingleton.*

Topology (Carlson's setting).

Theorem 4 (`isClosed_feasibilitySet`, `feasibilitySet_nonempty_of_compact`). *If each S_i is closed then S is closed. If moreover Ξ is compact and every finite subfamily is consistent (the finite intersection property, `FiniteConsistency`), then $S \neq \emptyset$.*

The compactness result is the abstract reason STE is well posed on Carlson's finite key spaces (discrete $\Rightarrow$ compact; `isClosed_feasibilitySet_of_discrete`) and on bounded closed convex constraint sets.

Convexity (Combettes's Hilbert-space setting).

Theorem 5 (`convex_feasibilitySet`). *If each S_i is convex then S (and every partial feasibility set) is convex.*

This is the geometric precondition for the projection algorithms of Bauschke and Borwein [1], Youla and Webb [4], a planned extension.

STE decryption skeleton (Carlson). A `CipherSystem` is a keyed family of injective encryption maps. Observing a ciphertext c induces the key property set $\{k \mid \exists m \in \text{Meaning},\ \text{enc}\, k\, m = c\}$; the feasible key set is the STE feasibility set of these.

Theorem 6 (`CipherSystem.fair_of_genuine`, `feasibleKeys_nonempty_of_genuine`). *Genuine traffic (ciphertexts that really are encryptions of meaningful messages under a true key k_0) is fair for k_0; hence the feasible key set is nonempty.*

Theorem 7 (`Unicity.eq_of_feasible`). *At unicity (feasible key set $= \{k_0\}$, i.e. ideal information) every feasible key is the true key: decryption succeeds by feasibility alone.*

4 Verification method

The environment used to author this work cannot download a Lean toolchain (egress policy). We therefore treat the GitHub Action as the trusted verifier: every push to `main` builds the `Ste` library with `leanprover/lean-action` against the pinned Mathlib cache. A result is considered established only once its build is green. This operationalizes the project's rule to distrust results that are not machine proven.

5 Future work

In increasing depth: Carlson's cipher counting (Lemma 4.1, Theorem 4.1); the reduction of all block ciphers to substitution ciphers (Theorem 5.2, Corollary 5.3); POCS convergence over Hilbert spaces; and Theorem 2.3, that property sets are AEP typical sets, placing information theory under STE. Targets and their open/proven status are tracked in the project `log/`.

References

[1] Heinz H. Bauschke and Jonathan M. Borwein. On projection algorithms for solving convex feasibility problems. *SIAM Review*, 38(3):367–426, 1996. doi: 10.1137/S0036144593251710.

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[3] Patrick L. Combettes. The foundations of set theoretic estimation. *Proceedings of the IEEE*, 81 (2):182–208, 1993. doi: 10.1109/5.214546. Retrieved from the author's page, `https://pcombet.math.ncsu.edu/proc.pdf`.

[4] Dan C. Youla and Heywood Webb. Image restoration by the method of convex projections: Part 1—theory. *IEEE Transactions on Medical Imaging*, 1(2):81–94, 1982. doi: 10.1109/TMI.1982.4307555.